Data Modelling Lab

DE25

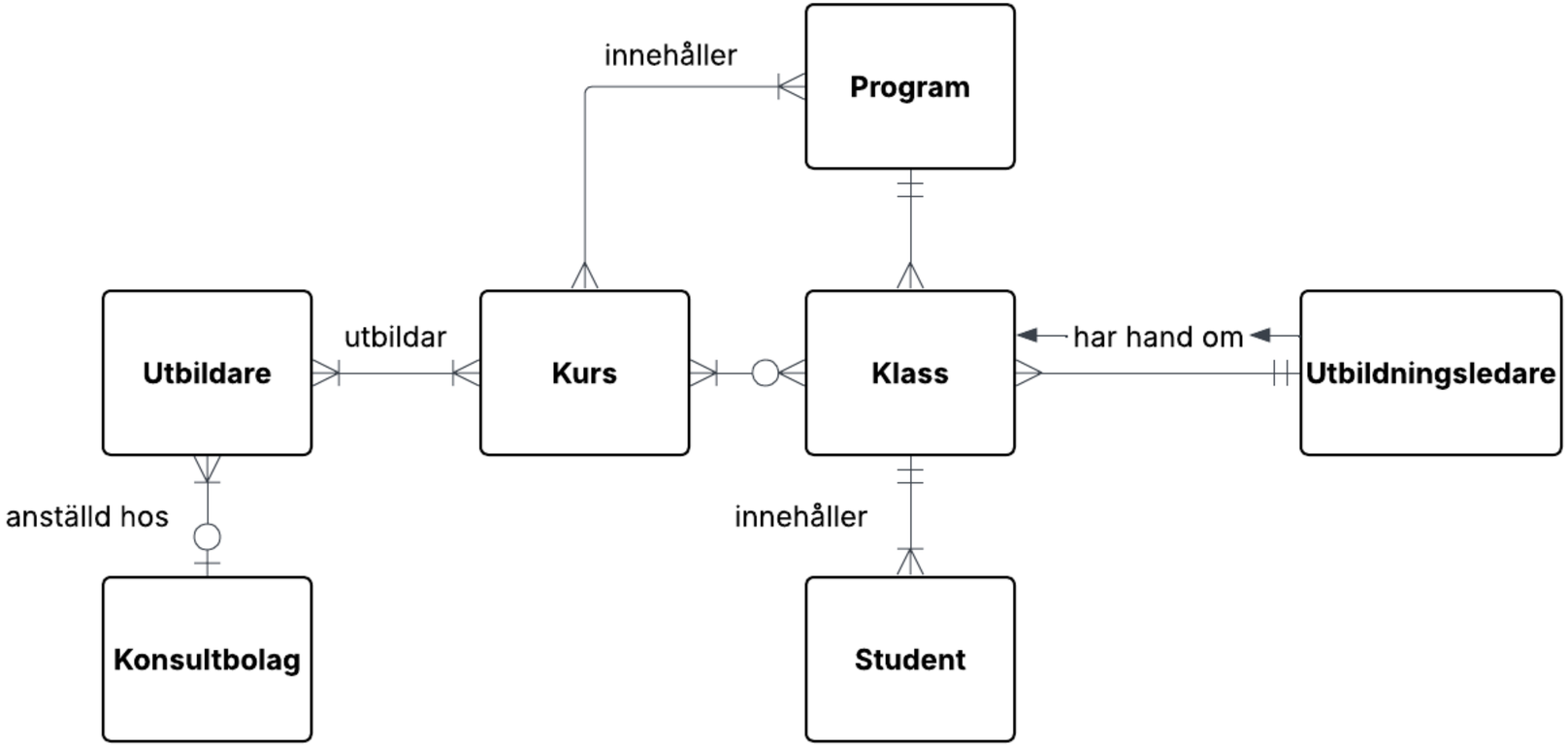
Purpose

- The purpose of this lab is to model a database for a school (YrkesCo)

Business requirements

- YrkesCo is in need of a database with the following information
  - Student first name, last name, personal identity number, email
  - Teachers can be consultants
  - Education manager personal information
  - Education manager is in charge of three classes
  - Courses with name, course code, points, description
  - Program consists of multiple courses
  - A program has three classes
  - Consultant company information

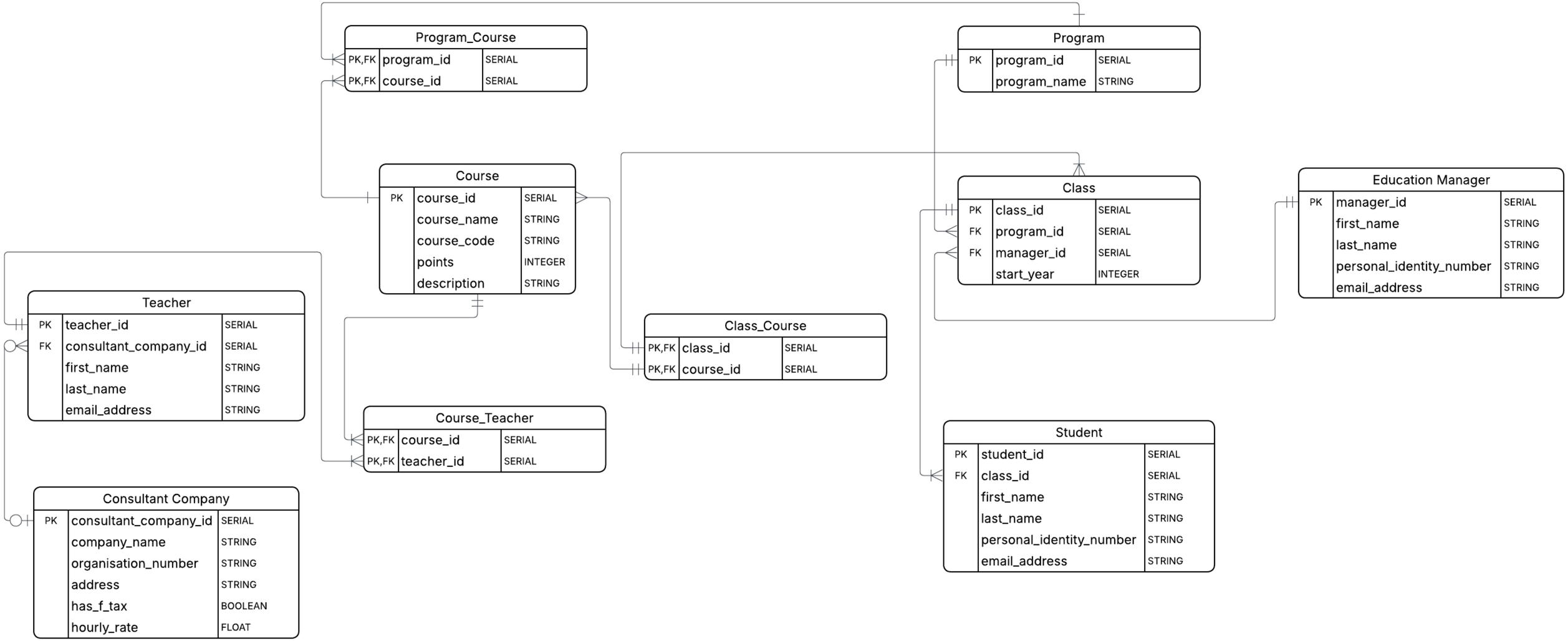
Conceptual ERD model



Conceptual ERD model

- The conceptual model is to convey the fundamental principles and basic functionality of the database
- The model has 7 core entities
- The Utbildare-Konsultbolag uses a zero-to-one relationship, which indicates that a teacher may or may not be a consultant
- The Kurs-Klass relationship uses a zero-to-many relationship, which means that there could exist a course that no class is currently enrolled in

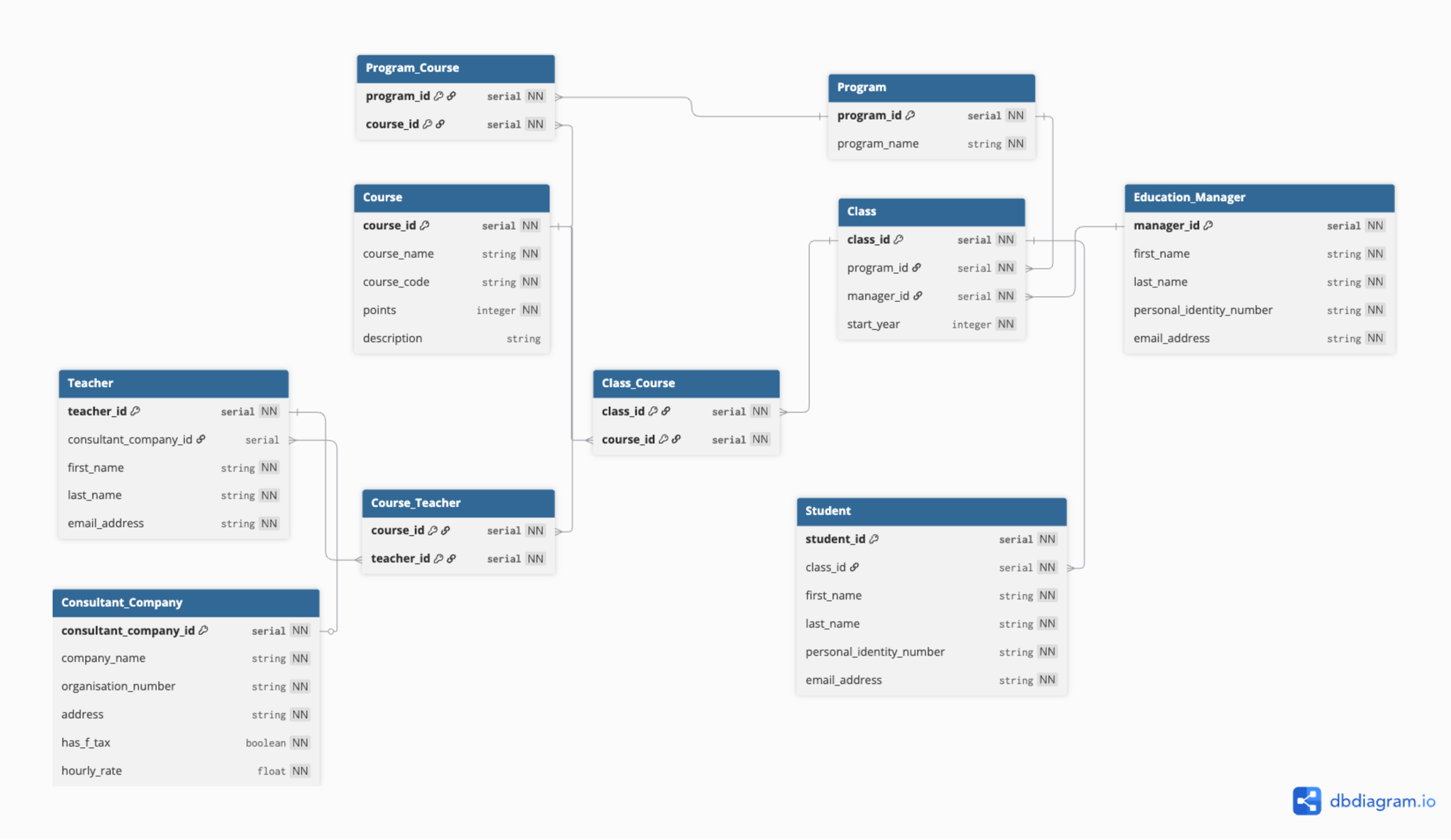
Logical ERD model



Logical ERD model

- The logical ERD model expands on the conceptual model
- Naming Primary and Foreign keys, describing how the tables relate to each other
- Describing the attributes of entities
- Asserting datatype to the attributes

Physical ERD model



Physical ERD model

- The physical model is the lowest level model
- Nullability is set
- Bridge tables are added, to solve the many-to-many relationships between entities

Implementation

- The database will be implemented in PostgreSQL by running it in a Docker container
- The database will get populated with mock data